



Name: Cohort/Batch: Date: Score:

APACHE ACTIVEMQ PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Messaging & Streaming

TOTAL: 10 QUESTIONS

Q1 What is Apache ActiveMQ and what problem in Messaging & Streaming does it solve?

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Q6 How does Apache ActiveMQ handle reliability and high availability?

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Q2 What are the core components or concepts in Apache ActiveMQ?

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Q7 What observability does Apache ActiveMQ provide out of the box?

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Q3 How does Apache ActiveMQ compare to its main alternatives?

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Q8 What are common performance bottlenecks in Apache ActiveMQ?

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Q4 How do you get started with Apache ActiveMQ for a new project?

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Q9 What security best practices apply when running Apache ActiveMQ?

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Q5 What configuration options most affect Apache ActiveMQ's behavior in production?

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Q10 What mistakes do teams commonly make when adopting Apache ActiveMQ?

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ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Apache ActiveMQ is a tool or platform

Mitigation

Apache ActiveMQ is a tool or platform that automates or simplifies a specific concern in the Messaging & Streaming domain, reducing manual effort and operational complexity for engineering teams.

A6 Apache ActiveMQ provides HA through leader election, replication, or stateless horizontal scaling

Observability

Apache ActiveMQ provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

A2 Apache ActiveMQ has key building blocks that

Primitives

Apache ActiveMQ has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

A7 Apache ActiveMQ exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces

Observability

Apache ActiveMQ exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

A3 Apache ActiveMQ trades off simplicity against feature richness differently than its alternatives

Hardening

Apache ActiveMQ trades off simplicity against feature richness differently than its alternatives. Choose Apache ActiveMQ when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

A8 Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches

Performance

Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

A4 Install Apache ActiveMQ via its package manager or binary release

Gotchas

Install Apache ActiveMQ via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

A9 Run Apache ActiveMQ with the principle of least privilege

Assessment

Run Apache ActiveMQ with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

A5 Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels

Hardening

Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

A10 Common pitfalls include skipping the documentation and learning the API by trial and error

Observability

Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.